

Multiple Choice

1. **Answer:** A

Options: A) Ne; B) CO₂; C) Ar; D) CH₄; E) O₂

Note: The correct answer is Ne because, according to Graham's law of effusion, the rate of effusion is inversely proportional to the square root of the molar mass of the gases, and helium's lower molar mass compared to neon results in a significantly higher effusion rate, consistent with the observed 5.33 times faster rate.

2. **Answer:** C

Options: A) In⁺, Sc³⁺ and Se²⁻; B) Ca²⁺, Ga³⁺ and In³⁺; C) Ca²⁺, Sc³⁺ and Se²⁻; D) In³⁺, Ga³⁺ and Se²⁻; E) Ca²⁺, In⁺ and Se²⁻

Note: Ca²⁺, Sc³⁺, and Se²⁻ possess a rare gas electronic configuration because they achieve the same electron configuration as the nearest noble gas by losing or gaining the appropriate number of electrons: Ca²⁺ loses two electrons to resemble argon, Sc³⁺ loses three electrons to also resemble argon, and Se²⁻ gains two electrons to mimic krypton.

3. **Answer:** A

Options: A) CO; B) CO₂; C) CH₄; D) BF₃; E) PCl₅

Note: The correct answer is CO because carbon monoxide has a linear molecular geometry due to the presence of a carbon atom bonded to an oxygen atom with a triple bond, resulting in an asymmetrical distribution of electron density around the molecule.

4. **Answer:** D

Options: A) 22.1 MeV; B) 23.3 MeV; C) 20.5 MeV; D) 21.2 MeV

Note: The correct answer of 21.2 MeV for the Coulombic barrier in the bombardment of ²⁰⁹Bi with alpha particles is derived from calculating the potential energy due to electrostatic repulsion between the alpha particle and the Bi nucleus.

4. **Answer:** A

Options: A) The atomic radius of hydrogen is five times the first Bohr radius; B) The atomic radius of hydrogen is four times the first Bohr radius; C) The atomic radius of hydrogen is half the first Bohr radius; D) The atomic radius of hydrogen is one tenth the first Bohr radius; E) The atomic radius of hydrogen is larger than the first Bohr radius

Note: The correct answer is based on the fact that the first Bohr radius, which is approximately 0.0529 nm, is smaller than the atomic radius of hydrogen due to the quantum mechanical model of the atom, where the Bohr model represents a simplified view of electron distribution that does not fully account for the electron cloud's probabilistic nature and energy levels in real atoms.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because decomposing 454 g of water requires 239.0 faradays of electricity, as the electrolysis of water produces two moles of hydrogen and one mole of oxygen, which requires a total of 4 faradays per mole of water.

7. **Answer:** True

Options: A) True; B) False

Note: Mass spectrometry data reveals distinct peaks corresponding to different isotopes of an element, allowing for the determination of their relative abundances and contributing to the understanding of variations in atomic mass.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the ideal gas equation is accurately represented by the relationship $PV = nRT$, where the ratio $(PV)/(RT)$ equals the number of moles (n), not 1, unless specifically stating that there is one mole of gas.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because while K^+ has a larger ionic radius than Na^+ , the ionic bond strength

9. **Answer:** True

Options: A) True; B) False

Note: The dipole moment of $HCOOH$ is calculated to be 1.41 D due to the molecular geometry and the polar bonds present in the molecule, which result in a net dipole moment when the $OCOH$ dihedral angle is 0° , aligning the dipoles effectively.

Long Form Response

11. **Answer:** The concept of half-life refers to the time required for half of a given quantity of a radioactive substance to decay. In this case, if 75% of the unstable substance decomposes in one hour, that implies 25% remains. Since 25% is half of 50%, we can infer that the substance must have reached this 25% level after one half-life. Consequently, if 75% decayed in one hour, it follows that the half-life of the substance is 30 minutes, as the remaining amount would halve from 50% to 25% in that time frame. Thus, the half-life of this radioactive substance is $1/2$ hour.
Note: correct because, given that 75% of the substance decays in one hour, the remaining 25% indicates that the substance must have halved from 50% to 25% in the subsequent 30 minutes, establishing the half-life as 30 minutes.
12. **Answer:** In interplanetary space, the relationship between pressure, density, and temperature can be described by the ideal gas law, which states that pressure (P) is propor-

tional to the product of density (ρ) and temperature (T), expressed mathematically as $P = kT$, where k is the Boltzmann constant. At a pressure of 10^{-16} torr, which is equivalent to 1.33×10^{-14} density gases in the vast expanses of interplanetary space.

Note: correct because it correctly applies the ideal gas law, $P = kT$, to relate pressure, density, and temperature.

End of Answer Key